Idemia Datachallenge (IADATA704): Face Occlusion Prediction

Group Name: Team GPU
Group Number: 21
Team Members: Oscar DE LA CRUZ, Rémi HAMON

Project repository: https://github.com/odelacruz-fierro/data-challenge-idemia.git

1 Introduction

In the context of the Data Challenge proposed by Idemia, the objective is to predict the percentage of facial occlusion in images (224x224 pixels). The major challenge lies in handling the training data imbalance (a low proportion of the training database contain highly occluded faces) and the gender fairness requirement imposed by the evaluation metric.

In this report, we present our solution based on multi-task learning and data augmentaion, designed to extract robust features while minimizing the demographic bias penalty.

2 Exploratory Data Analysis

The exploratory data analysis (EDA) performed on the train database revealed that only a low proportion of the faces show a high occlusion. This is verified by Fig.1, where we can observe a right-skewed distribution: the samples are more concentrated on the lower values than in the extreme high values, whici drag the tail towards the right side.

From a gender perspective, it was found that the train database contains 67.6 % of men and 32.4 % of women (see Fig.2).

In order to understand what different occlusion levels looked-like, images were sampled out from the train database using different occlusion intervals, *i.e.* $Low \in [0.1, 0.3]$, $Medium \in [0.31, 0.4]$ and $High \in [0.7, 1.0]$. These intervals were defined based on the test database occlusion distribution provided with the Data Challenge (see Fig.3). The results of this sampling process allow to make some interesting observations:

- *low* occluded images trend to provide a global visibility of the face forehead, eyes, nose, chicks and chin. It's worth noting that the neck is not visible.
- in *medium* occluded images par of the faces are covered either with the hair, a hat, hand, fingers and even physical objects.
- Finally in *high* occluded images faces seem to be mostly visible, however a blurry / noised effect was noticed. It's quite interesting that there were only 5 images in this interval. These can defnitely be considered as outliers.

Figure 4 provides simples for each interval. The above-presented findings will be tackled during in section 4.2.

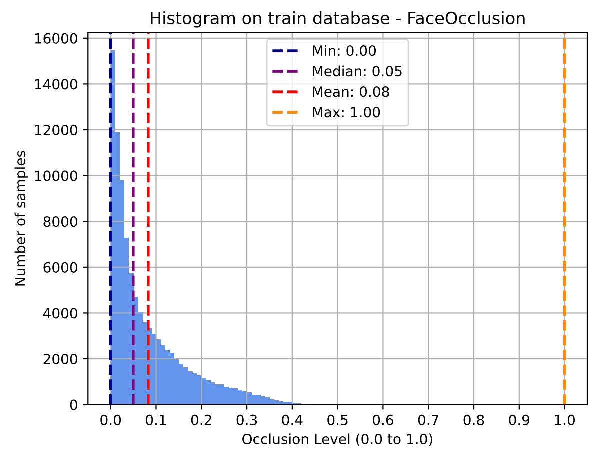

Figure 1: Distribution of occlusion across the training database.

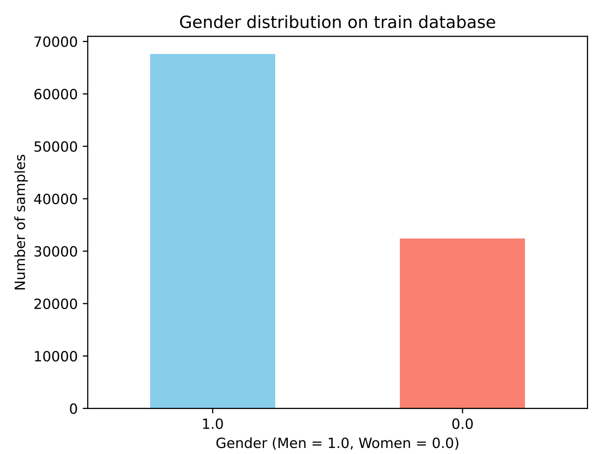

Figure 2: Distribution of gender across the training dataset: Men = 67.6 % and Women = 32.4 %.

3 Methodology & Architecture

TO DO ...

3.1 Milti-Task Learning (MLT)

TO DO ...

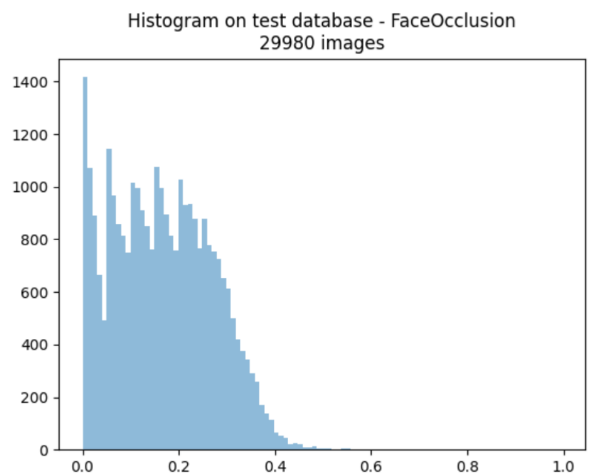

Figure 3: Distribution of occlusion across the test database.

3.2 Backbone architectures

TO DO ...

4 Training strategy

TO DO ...

4.1 Custom loss

TO DO ...

4.2 Data augmentation

Interesting fact, in most of the pictures the face covers most of the frame.

5 Results

TO DO ...

6 Limitations and tradeoffs

7 Conclusion

TO DO ...

Figure 4: Random samples for three occlusion intervals: Low $\in [0.1, 0.3]$, Medium $\in [0.31, 0.4]$ and High $\in [0.7, 1.0]$.